Nicolás Muñoz García

BSc in Physics (2025–2029)
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Education

- **Universidad de Salamanca (USAL)** — *2025 – Present*
  BSc in Physics – expected June 2029 — Salamanca, Spain
- **Upper Secondary Education (Bachillerato), Science and Technology track** — *2023 – 2025*
  University entrance examination (PAU): 12.501/14 — Average: 9.06/10
- **Universidad de Salamanca – Additional Training** — *September–October 2026*
  Artificial Neural Networks: Current Models and the Deep Learning Paradigm
  30 hours | 2 ECTS credits
  - Successfully completed a course covering transfer learning with deep convolutional networks, U-Net semantic segmentation, generative models (VAEs and GANs), recurrent networks, natural language processing with LLMs and an introduction to latent diffusion models.

Projects

- **PaperTok – papertok.app** — *2025 – Present*
  Founder & Developer — mugar123/papertok
  - Founded and grew an open-source research discovery platform to 700 users, bringing scientific papers from arXiv, OpenAlex, PubMed and other sources into a personalized, scrollable feed.
  - Developed a university-library integration that enables students to link their institution and access paywalled papers discovered on PaperTok through their university's subscriptions, connecting research discovery with access to full text.
  - Develop personalized recommendations that combine semantic embeddings with readers' interests and interactions, helping users find relevant work across disciplines and beyond their usual searches.
  - Work with a team of developers to bring discovery and reading into one platform, with citation maps, shareable reading lists and AI-assisted explanations tailored to different levels of scientific knowledge.
  - Secured support through Claude for Startups, Codex for OSS and individual sponsors to advance the platform's development.

  Technologies: TypeScript, React, Vite, Cloudflare Workers, D1, Firebase, semantic embeddings and scientific APIs.
- **Automated LaTeX Lecture Notes Pipeline** — *2025 – Present*
  Personal project – Mathematics & Physics
  - Built an AI-assisted pipeline to convert class notes into structured LaTeX documents, automating a previously manual transcription and typesetting workflow.
  - Designed a multi-model workflow using OpenClaw, with Claude Opus 5.5 orchestrating transcription tasks across low-cost open-source models and reconstructing lecture figures and diagrams as LaTeX graphics.
  - Developed a reusable LaTeX template to give generated notes consistent formatting, mathematical notation and document structure, producing course materials shared with classmates.

  Technologies: OpenClaw, Claude Opus 5.5, open-source language models, LaTeX.

Technical Skills and Interests

Programming: Python, JavaScript/TypeScript, MATLAB
Software development: Frontend and backend development, API integration, version control with Git and GitHub
AI and automation: AI-assisted development and automation; proficient in Claude Code, Codex and Cursor
Cloud and databases: Cloud computing, database management and application deployment
Scientific tools: Mathematica, PHYWE measure, LabVIEW, LaTeX (advanced), scientific computing and technical documentation
Languages: Spanish (native); English (B2, certified by Escuela Oficial de Idiomas, Spain's public system for language teaching and official certification)
Relevant coursework: Mathematical analysis, linear algebra, differential equations, classical mechanics, electromagnetism, thermodynamics and experimental physics
Areas of interest: Computational physics, machine learning and astrophysics

Extracurricular Activities

- **Member**, Supernova (student association, Universidad de Salamanca) — *2025 – Present*
- **Participant**, astronomy outreach activities organised by Supernova — *2025 – 2026*

Achievements

- **PaperTok**, accepted into Claude for Startups and Codex for OSS.
- **Participant**, PLANCKS 2027 – international theoretical physics competition — *Nov. 2026*
- **PAU score of 12.501/14**, Spanish university entrance examination — *2025*
- **Participant**, Spanish Physics Olympiad (Olimpiada de Física), two consecutive editions — *2024, 2025*